

REQUIREMENTS SPECIFICATION FOR

Visual-Inertial Data Collection Platform and Sensor Fusion Benchmarking

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PROJECT MANAGEMENT

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Goal : Build a drone-mounted sensor platform for collecting ground-truth-referenced datasets, then evaluate state-of-the-art fusion and VIO algorithms against OptiTrack.

Request :

Design :

Design and assemble a modular sensor platform integrating the following sensors (see [here](#)) :

- Camera : Intel RealSense D435i (<https://realsenseai.com/products/depth-camera-d435i/>) and/or Olive Robotics OliveVision V1
- IMUs (2–3 units) : Olive Robotics OliveSense X1 Pro
- Optical flow : Holybro H-Flow
- Onboard compute : Jetson Orin Nano (or equivalent), running ROS 2 Humble
- Mounting points for OptiTrack markers
- Room for future sensor expansion (extra IMUs, depth cameras, etc.)

Build a ROS 2 Architecture :

Build a ROS 2 data collection pipeline. All of the sensors need to be brought up as ROS 2 nodes with proper time synchronization. Initial flights and recording will be done in CRTA under the OptiTrack system, which will provide ground-truth pose. Record everything as rosbag2 with synchronized timestamps across all sensors.

sensor fusion algorithms :

Process the recorded datasets using sensor fusion algorithms. Implement and compare classical and modern filters in ROS 2, EKF, UKF, and InEKF (Invariant EKF). Other variants (e.g., MSCKF) are welcome.

Benchmark visual-inertial odometry / SLAM frameworks :

Benchmark visual-inertial odometry / SLAM frameworks on the same datasets. Target candidates : OpenVINS, VINS-Fusion, ORB-SLAM3, and others worth comparing, all run within ROS 2.

Quantitatively evaluate all estimated trajectories :

Quantitatively evaluate all estimated trajectories against OptiTrack ground truth using the evo Python package (<https://github.com/MichaelGrupp/evo>) — APE, RPE, trajectory alignment, plots.

Deliverables :

Working hardware platform with full ROS 2 stack, recorded rosbag2 datasets with synchronized ground truth, fusion algorithm implementations, and a comparative report of VIO algorithm performance.

Rendering Goal :

One design for a modular sensor platform and an other design to integrate the platform on a drone (UAV or else).

Build a ROS 2 Architecture :

Build a ROS 2 data collection pipeline. All of the sensors need to be brought up as ROS 2 nodes with proper time synchronization. Initial flights and recording will be done in CRTA under the OptiTrack system, which will provide ground-truth pose. Record everything as rosbag2 with synchronized timestamps across all sensors.

Sensor fusion algorithms :

Process the recorded datasets using sensor fusion algorithms. Implement and compare classical and modern filters in ROS 2, EKF, and UKF.

Benchmark visual-inertial odometry :

Benchmark visual-inertial odometry : OpenVINS and IsaacRosVisualSlam run within ROS 2.

Quantitatively evaluate all estimated trajectories :

Quantitatively evaluate all estimated trajectories against OptiTrack ground truth using the evo Python package — APE, RPE, trajectory alignment, plots.

Deliverables :

Working hardware platform with full ROS 2 stack, recorded rosbag2 datasets with synchronized ground truth, fusion algorithm implementations, and a comparative report of VIO algorithm performance.